

MS4215: Lab 8

Logistic Regression: Coronary Heart Disease

Sinéad Moylett

Introduction

You have seen this dataset before. In Lab 3, `heart_study.csv` was used to model systolic blood pressure (a continuous outcome) using linear regression. In this lab the outcome switches to `CHD`, a binary indicator of coronary heart disease diagnosis, and the method switches to logistic regression.

The dataset contains 198 patients from a cardiovascular study. Four patients with missing cholesterol values are removed in the setup chunk, leaving 194 observations. The variables are:

Variable	Description
RANDID	Patient identifier
GENDER	Sex (0 = Female, 1 = Male)
TOTCHOL	Total cholesterol (mg/dL)
AGE	Age in years
SYSBP	Systolic blood pressure (mmHg)
CURSMOKE	Current smoker (0 = No, 1 = Yes)
CHD	CHD diagnosis (0 = No, 1 = Yes)

The lab has two parts. **Part A** explores the binary outcome and fits simple logistic regression models. **Part B** builds a multiple logistic regression model, compares nested models using the likelihood ratio test, and uses the final model to predict CHD probability for new patients.

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("tidyverse")
library(tidyverse)

# In the line of code below, you will need to change the location to where you
# have saved the file
heart <- read_csv("data/heart.csv") |>
  na.omit() |>
  mutate(
    GENDER.f = factor(GENDER, levels = c(0, 1), labels = c("female", "male")),
    CURSMOKE.f = factor(CURSMOKE, levels = c(0, 1), labels = c("non-smoker", "smoker")),
    CHD.f = factor(CHD, levels = c(0, 1), labels = c("No CHD", "CHD"))
  )
```

Questions

Part A: Exploratory analysis and simple logistic regression

- (i) Explore the data

Use `summary()` to inspect the dataset. How many patients were diagnosed with CHD? What proportion of the sample has CHD?

Then produce bar charts showing the count of CHD diagnoses by `GENDER.f` and by `CURSMOKE.f`. Use `fill = CHD.f` and `position = "dodge"`. Comment on whether CHD appears to differ by gender or smoking status.

- (ii) Odds ratio from a 2×2 table

Build a CHD × GENDER.f crosstab using `table()`. Use `prop.table(..., 1)` to display row proportions.

From the table counts, compute the odds ratio (OR) manually using the cross-product formula:

$$\text{OR} = \frac{AD}{BC}$$

Then apply the Woolf formula to compute a 95% confidence interval:

$$95\% \text{ CI} = \exp\left(\ln(\text{OR}) \pm 1.96\sqrt{\frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D}}\right)$$

Interpret the OR. Does the confidence interval include 1?

- (iii) Simple logistic regression: CHD ~ GENDER.f

Fit a logistic regression model for CHD with gender as the only predictor. Use `glm()` with `family = binomial`. Use `summary()` to inspect the output.

- What is the reference category for `GENDER.f`?
- Interpret the intercept on the log-odds scale. Verify it against the table from (ii).
- Interpret the `GENDER.fmale` coefficient on the log-odds scale.
- Use `exp(coef(m))` and `exp(confint(m))` to obtain the OR and 95% profile likelihood CI. Compare the OR to the table-based OR from (ii).

Part B: Multiple logistic regression and model comparison

- (iv) Multiple logistic regression

Fit the following two models:

- `m1: CHD ~ AGE + GENDER.f`
- `m2: CHD ~ AGE + TOTCHOL + GENDER.f + CURSMOKE.f`

For `m2`, use `exp(coef(m2))` and `round(exp(confint(m2)), 3)` to obtain adjusted ORs and 95% CIs for all predictors. Identify which predictors are statistically significant at the 5% level (use the Wald z-values and p-values from `summary()`).

Interpret the adjusted OR for `AGE`: what does it mean for a 10-year increase in age?

- (v) Model comparison: likelihood ratio test (LRT) and AIC

Use `anova(m1, m2, test = "LRT")` to test whether adding `TOTCHOL` and `CURSMOKE.f` to `m1` significantly improves model fit.

- Report the LRT statistic, degrees of freedom, and p-value.
- What is the conclusion?

Use `AIC(m1, m2)` to compare both models. Does AIC agree with the LRT?

(vi) Fitted probabilities

Using `m2`, compute the predicted probability of CHD for the following two patient profiles:

- A 55-year-old male smoker with total cholesterol of 280 mg/dL
- A 55-year-old female non-smoker with total cholesterol of 220 mg/dL

Use `predict(m2, newdata = ..., type = "response")`. Comment on what the model tells us about CHD risk for these two patients.